June 14 — Login Page Development

Activity: Login Page Development

On June 14, the Login Page for the Real Time Groundwater Resource Evaluation System is developed. This serves as the secure entry point for authorized users such as hydrogeologists, field engineers, and government officials to access the DWLR data monitoring dashboard.

What is Login Page Development?

Login Page Development involves building a secure authentication interface that verifies the identity of users before granting access to sensitive groundwater monitoring data collected from Digital Water Level Recorders (DWLRs). Given the critical nature of real-time groundwater data, the login system must be robust and role-aware.

Key Tasks Involved:

- Design the login form layout using React components tailored to the groundwater monitoring system's UI theme.
- Implement username/email and password fields with real-time validation and error feedback.
- Connect the login form to the Spring Boot backend Authentication API for credential verification.
- Integrate JWT (JSON Web Token) based session management to maintain secure user sessions.
- Implement role-based access control (RBAC) — Admin, Field Engineer, and Viewer roles for DWLR data access.
- Handle login errors: invalid credentials, account locked, and server unavailability.
- Apply responsive design for field use on mobile devices and tablets near DWLR stations.

Expected Deliverable:

Login Module — A fully functional, secure login page integrated with the groundwater evaluation system's backend, enabling authenticated access to real-time DWLR data dashboards and reports.

Technology	Purpose
React.js	Frontend UI for the login interface
Axios / Fetch API	API calls to Spring Boot authentication endpoint
JWT	Secure token-based session management for DWLR data access
React Router	Redirect to Groundwater Dashboard after login
CSS / Tailwind	Responsive styling for field and office use
Spring Boot (Backend)	Validate credentials and issue JWT tokens

June 15 — Registration Page Development

Activity: Registration Page Development

On June 15, the Registration Page is developed to allow new authorized users to create accounts within the Real Time Groundwater Resource Evaluation system. This page ensures that only verified personnel can register and gain access to DWLR sensor data and groundwater analytics.

What is Registration Page Development?

Registration Page Development involves building a form-based interface where new system users — such as district-level groundwater officers, monitoring engineers, or administrators — can sign up by providing their details. The registration data is validated on the frontend and securely sent to the backend for account creation and role assignment.

Key Tasks Involved:

- Design the registration form with fields: Full Name, Email, Employee ID, Department, Password, Confirm Password, and Role.
- Implement client-side validation for all fields — email format, password strength, and ID format specific to groundwater department.
- Connect to the Spring Boot Registration API endpoint to securely submit new user data.
- Implement role selection during registration (Admin / Field Engineer / Viewer) with backend role mapping.
- Show success confirmation and redirect to Login Page after successful account creation.
- Handle backend errors: duplicate email, existing Employee ID, and unauthorized domain registrations.
- Maintain consistent UI design with the Login Page for a seamless user experience across the system.

Expected Deliverable:

Registration Module — A complete, validated registration page that enables new groundwater monitoring personnel to create accounts, assigns appropriate roles for DWLR data access, and integrates with the Login Module for a secure end-to-end authentication flow.

Technology	Purpose
React.js	Component-based registration form UI
Axios / Fetch API	POST request to Spring Boot registration endpoint
React Hook Form / useState	Manage form state, validation, and role selection
CSS / Tailwind	Consistent styling matching the Login Page
React Router	Redirect to Login after successful registration
Spring Boot (Backend)	Create user account and assign role in the database